

Speech and Audio Processing

## 02 - speech production

Shinnosuke Takamichi / 高道 慎之介



Takamichi Laboratory  
慶應義塾大学 高道研究室

# Class schedule (tentative)

|           |                |                                       |                                                               |
|-----------|----------------|---------------------------------------|---------------------------------------------------------------|
| 01        | Oct. 02        | Introduction                          | Learn from basic to state-of-the-art about speech processing. |
| <b>02</b> | <b>Oct. 09</b> | <b>Speech production / 音声生成</b>       |                                                               |
| 03        | Oct. 16        | Speech recognition / 音声認識             |                                                               |
| 04        | Oct. 23        | Speech synthesis / 音声合成               |                                                               |
| 05        | Oct. 30        | Self-supervised learning / 自己教師あり学習   |                                                               |
| 06        | Nov. 6         | Foundation model / 基盤モデル              |                                                               |
| 07        | Nov. 13        | Sound recognition/synthesis / 環境音認識合成 |                                                               |
| 08        | Nov. 27        | Advanced topics / 発展的課題               |                                                               |
| 09        | Dec. 04        | Advanced topics / 発展的課題               |                                                               |
|           |                |                                       |                                                               |
| 10        | Dec. 11        | Speech system building / 音システム構築      | Learn how to build your speech system.                        |
| 11        | Dec. 18        | Speech system building / 音システム構築      |                                                               |
| 12        | Dec. 25        | Speech system building / 音システム構築      |                                                               |
| 13        | Jan. 08        | Speech system building / 音システム構築      |                                                               |
| 14        | Jan. 22        | <b>Final presentation / 最終発表</b>      |                                                               |

# About this course

---

- Language
  - Slides in English (w/ partial translation to Japanese, e.g., named entity)
  - Talk in **English** (w/ simultaneous translation to Japanese.)
  - Either En or Ja is fine for your question.
- Grades
  - 50 / 100 ... final presentation
  - 50 / 100 ... final report
- Room
  - 14-204

# Final presentation & reports / 最終発表&レポート

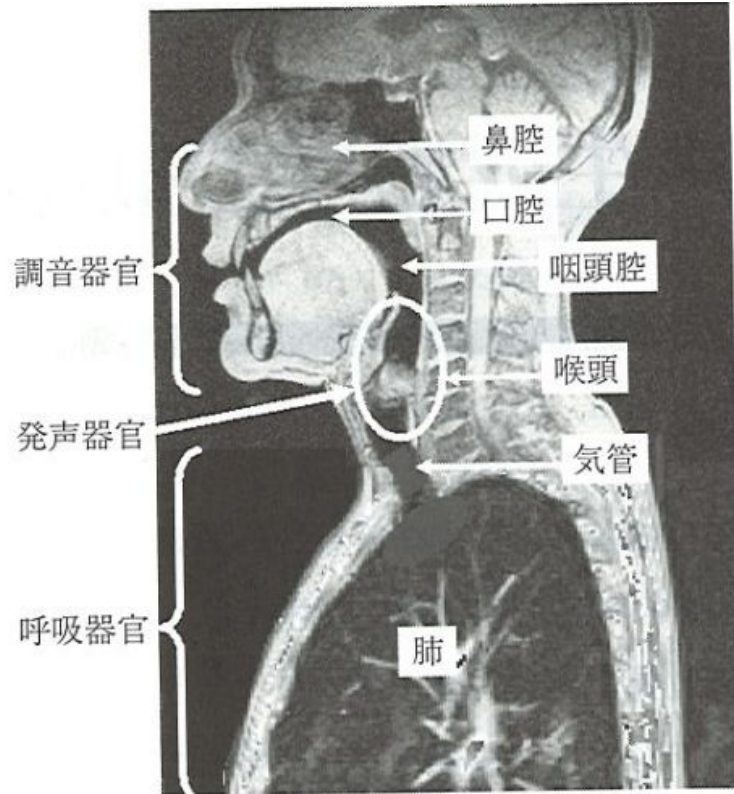
---

- Presentation
  - Oral presentation (duration: [TBA])
  - Talk about purpose, system details, and evaluation of your system.
  - (Advanced:) submit to domestic or international conferences.
    - Last year: 1 team accepted for Interspeech. 4 submitted to domestic conf.
- Report
  - 2 page, no specific format. Describe purpose, systems, and evaluation.
- Language
  - Slides in English.
  - Talk in Japanese or English. (choose one you prefer.)
    - I will drive {Ja, En}-to-{En, Ja} simultaneous translation during presentation.
  - Reports in Japanese or English. (choose one you prefer.)

# Today's goal

---

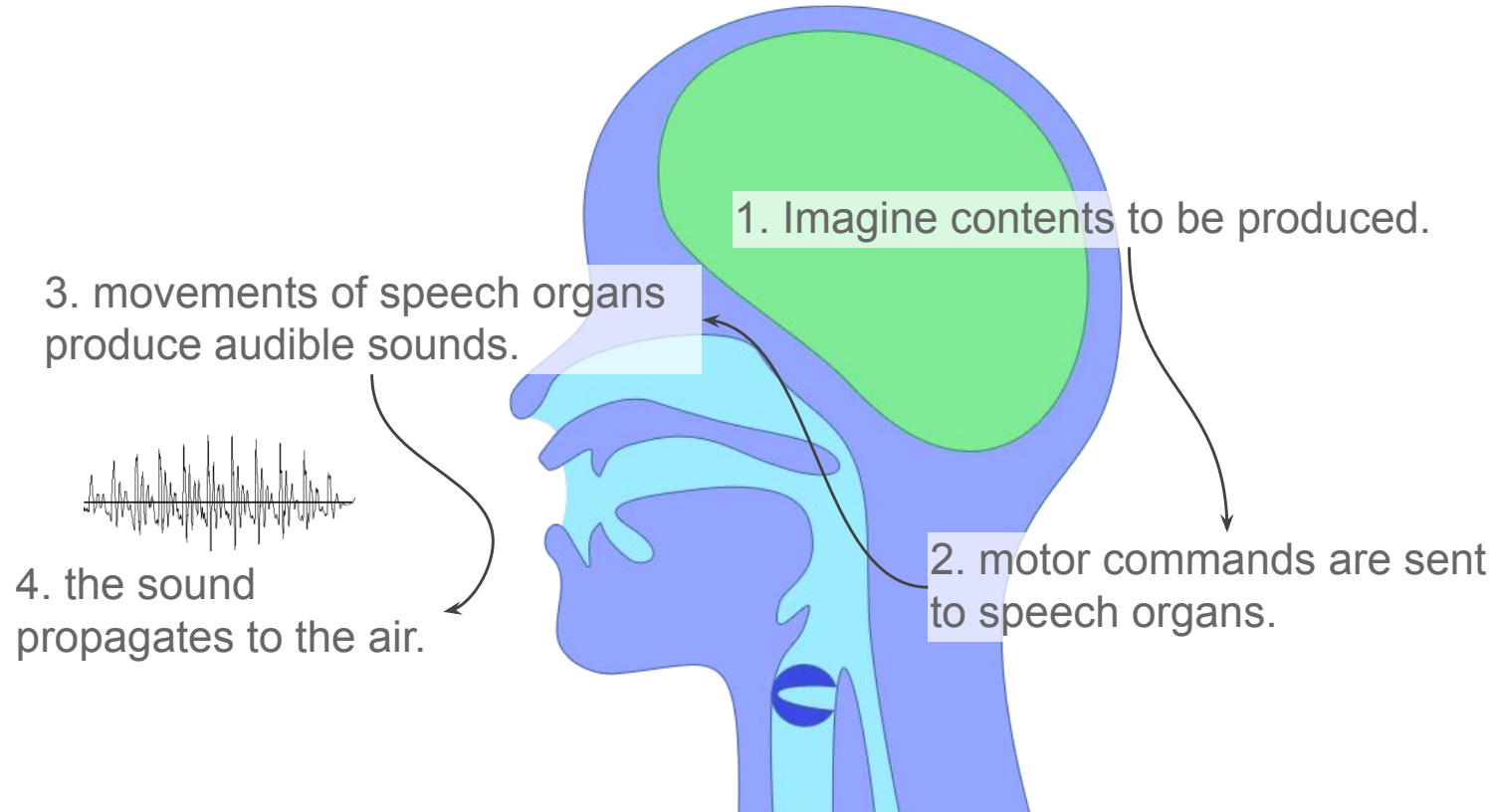
## What happens in our body when producing speech



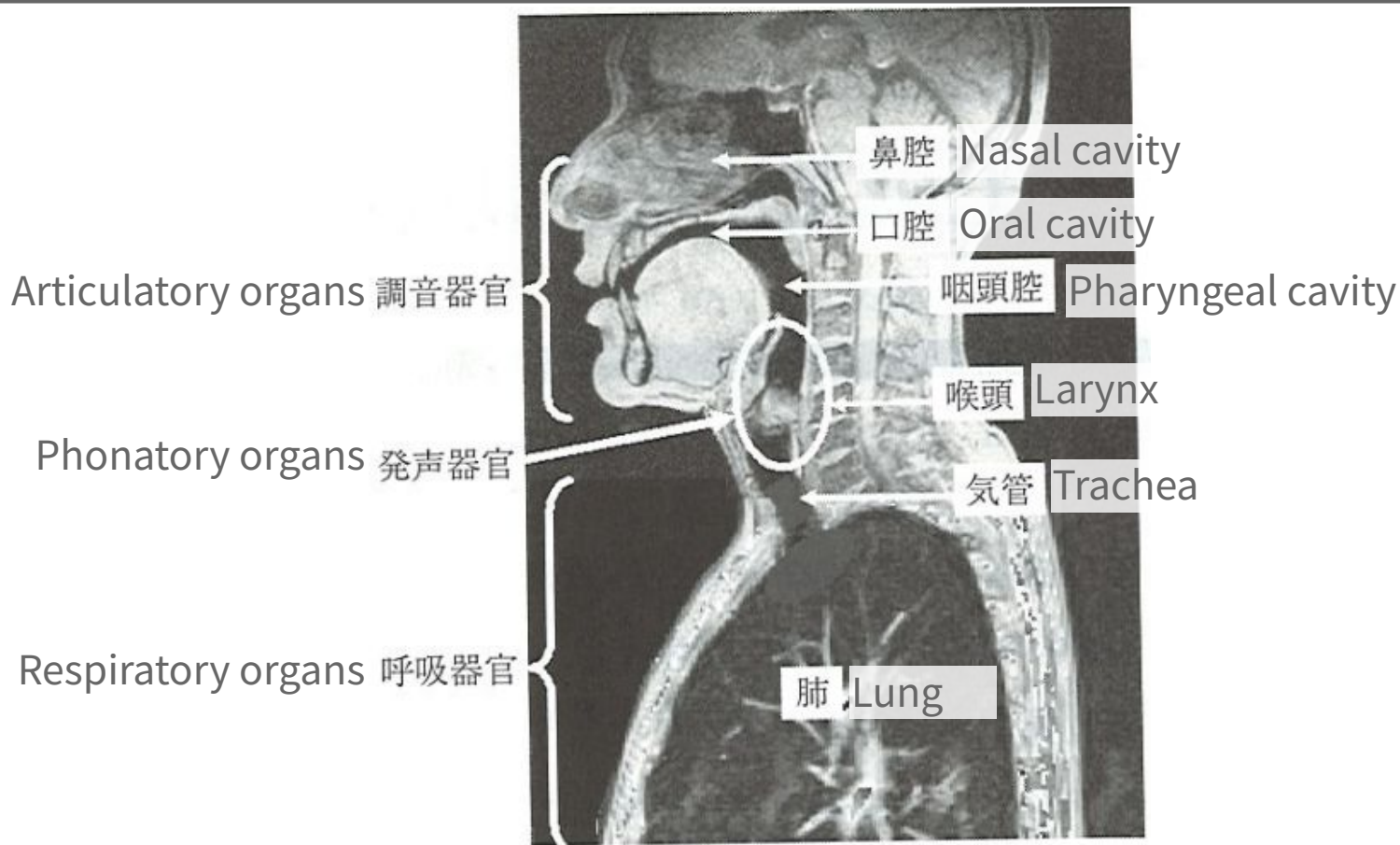
# Speech production / 音声生成

# Speech = human voice (sounds produced via human organs)

---

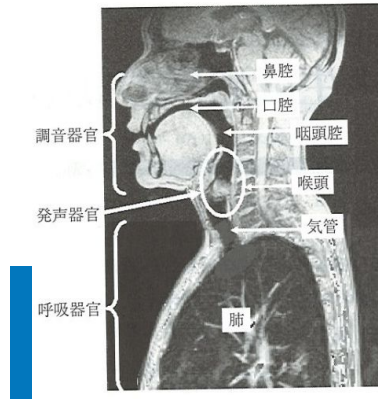


# Speech organs / 音声器官





# Respiratory organs / 呼吸器官



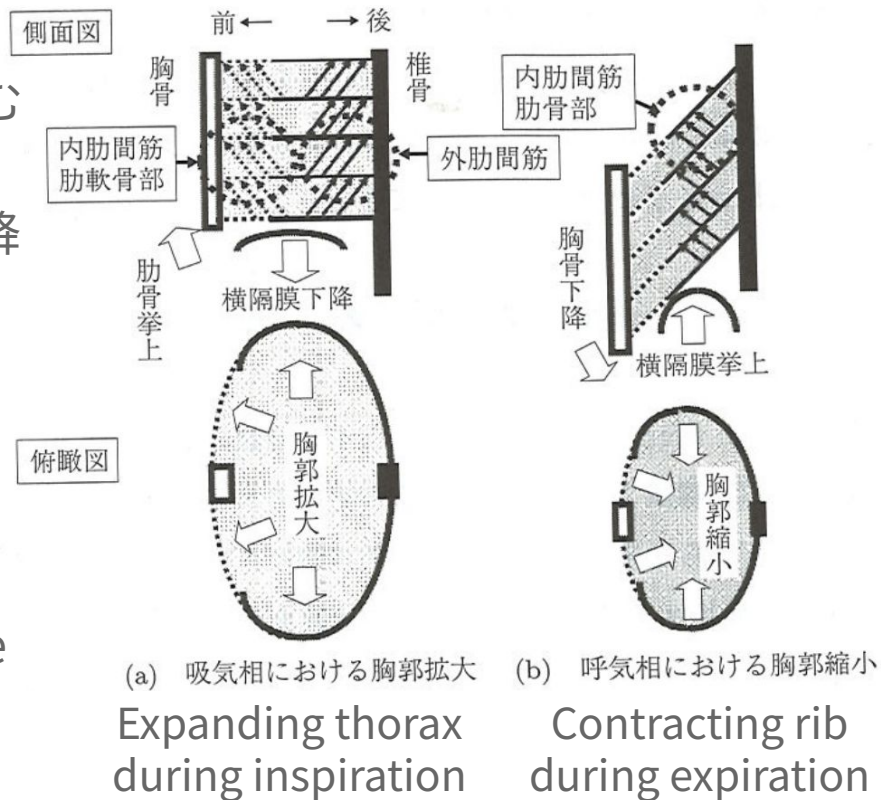
# Respiratory organs / 呼吸器官

- **Inspiration (吸気)**

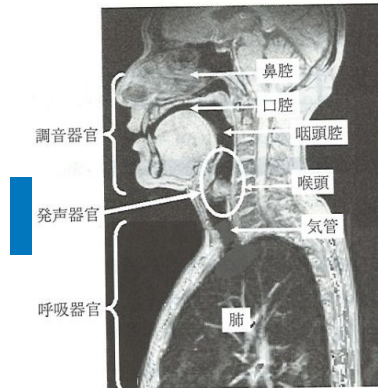
- Sending air to lungs / 肺へ吸い込む
- The diaphragm descends and the thorax expands. / 横隔膜の下降と胸郭の拡大

- **Expiration (呼気)**

- Expelling air from the lungs
- The diaphragm rises and the internal intercostal muscles of the rib portion contract. / 横隔膜の挙上と内肋間筋の肋骨部の収縮



# Phonatory organs / 発声器官



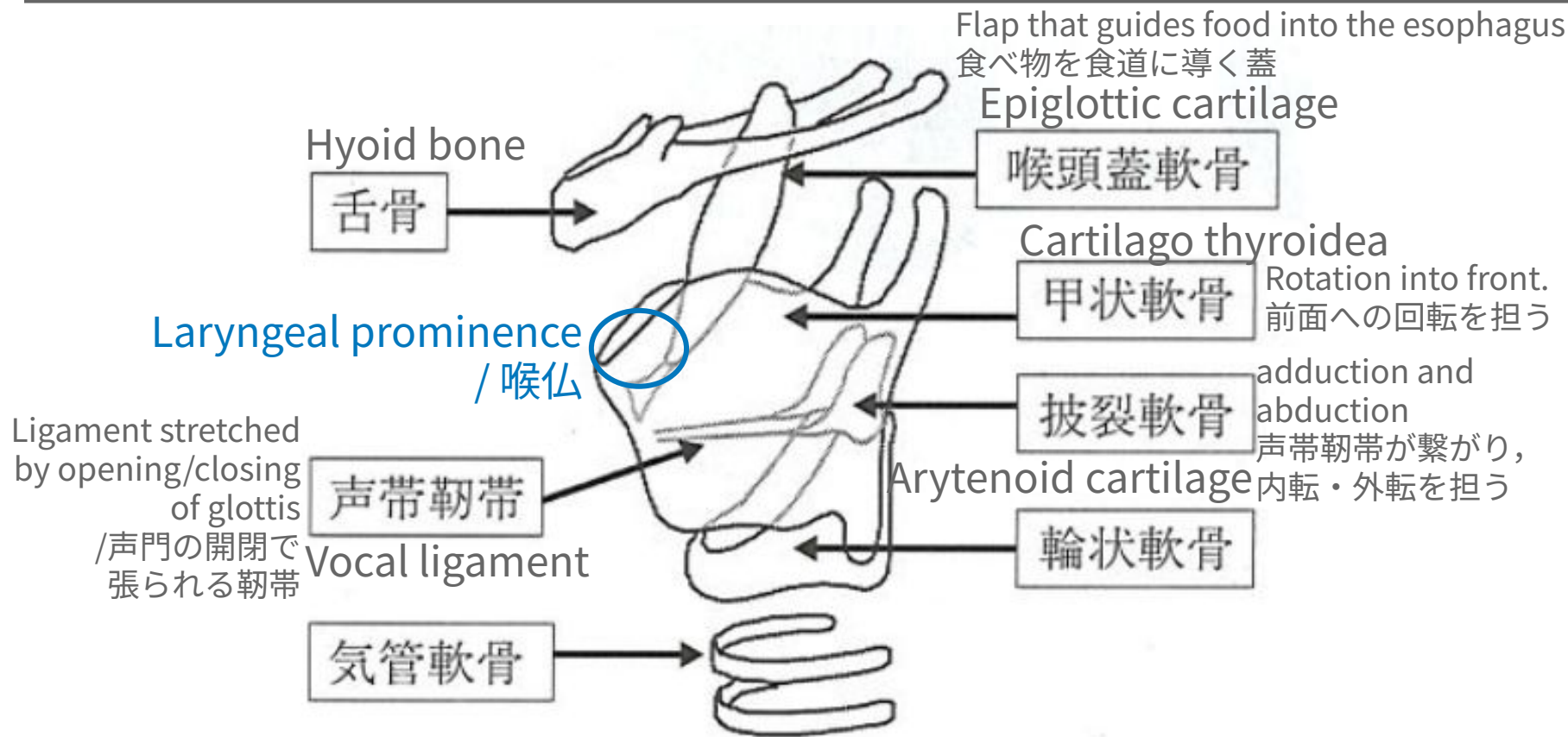
# Phonatory organs / 発声器官

---

Making voicing/unvoicing (w/ or w/o vibration) sounds by this air valve.



# Phonatory organs / 発声器官



# Physical model around larynx / 喉頭近辺の物理モデル



View from front

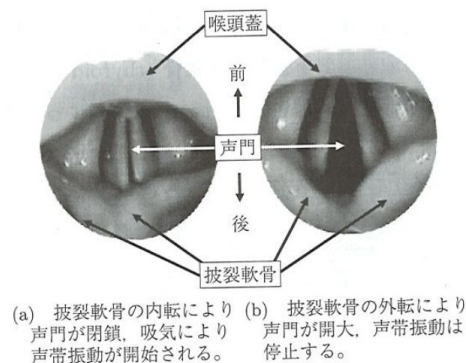
View from back



# Generation of burst and turbulent noise by abduction of the arytenoid cartilages / 披裂軟骨の外転による破裂音源・乱流雑音源の生成

## • Arytenoid cartilage / 披裂軟骨

- Glottis is closed by adduction / 内転で声門が閉鎖
- Glottis is open by abduction / 外転で声門が開大



## • Semi-periodic open-close movements / 準周期的な開閉運動

- Adduction closes glottis / 内転により声門が閉鎖
- Air pressure in subglottis becomes greater than supraglottis / 肺からの呼気により声門下の圧力が、声門上より大きくなる
- The pressure difference causes the glottis to open, allowing air to flow above the glottis / 圧力差により声門が開き、少量の空気が声門上へ

# Fundamental frequency / 基本周波数

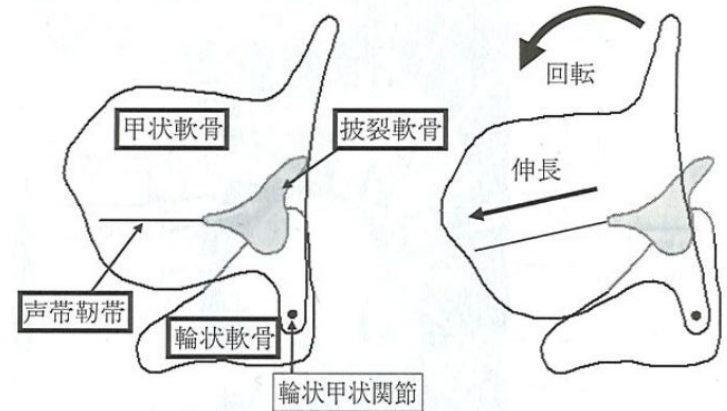
---

- Inverse of the period of open-close movements / 開閉運動周期の逆数
  - Roughly equal to “pitch” / いわゆる”声の高さ”
  - Around 100 Hz for adult males, 200 Hz for adult females / 男性の場合は 100 Hz前後, 成人女性の場合は 200 Hz前後
  - The symbol  $F_0$  (エフゼロ) is used.
- Note
  - The fundamental frequency does not always exist. / 基本周波数は常に存在するわけではない.
  - In the case of turbulent noise, the fundamental frequency is absent. / 後述する破裂音源・乱流雑音源では, 基本周波数は存在しない.
  - It is treated as 0 Hz in computers but does not mean 0Hz in physics. / コンピュータでは 0 Hz と扱うこともあるが, 物理的に 0 Hz ではない

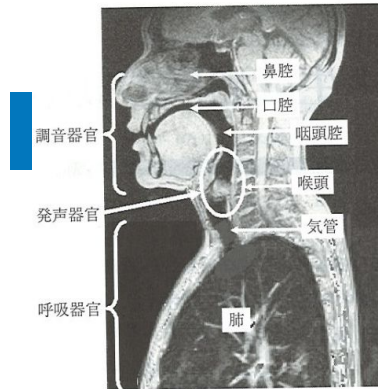


# How we control the fundamental frequency? / 基本周波数をどうやって変えているのか？

- Rotation of cartilago thyroidea (輪状軟骨) and arytenoid cartilage (甲状軟骨)
  - Angle by two becomes larger (moving front side)
    - Vocal ligament stretched tightly, increasing the stiffness of vocal folds. / 声帯靱帯が強く張られ、声帯の剛性が上がる
    - The natural frequency of the vocal folds becomes higher → the fundamental frequency increases. / 声帯の固有周波数が高くなる = 基本周波数が上がる
  - In the opposite process (= loose), the fundamental frequency decreases. / 角度が小さくなると基本周波数は下がる



# Articulatory organs / 調音器官



# Articulatory organs / 調音器官

日本語の「ワ」と「ファ」

「ワ」と「ファ」の発音のメカニズムを、リアルタイムMRI映像を用いて解説します。舌の位置や口の形状の変化が、音の違いを生み出す原因となります。



耳鳴りと耳鳴りマイクログラフ



東京話の「ワ」「ファ」の子音群





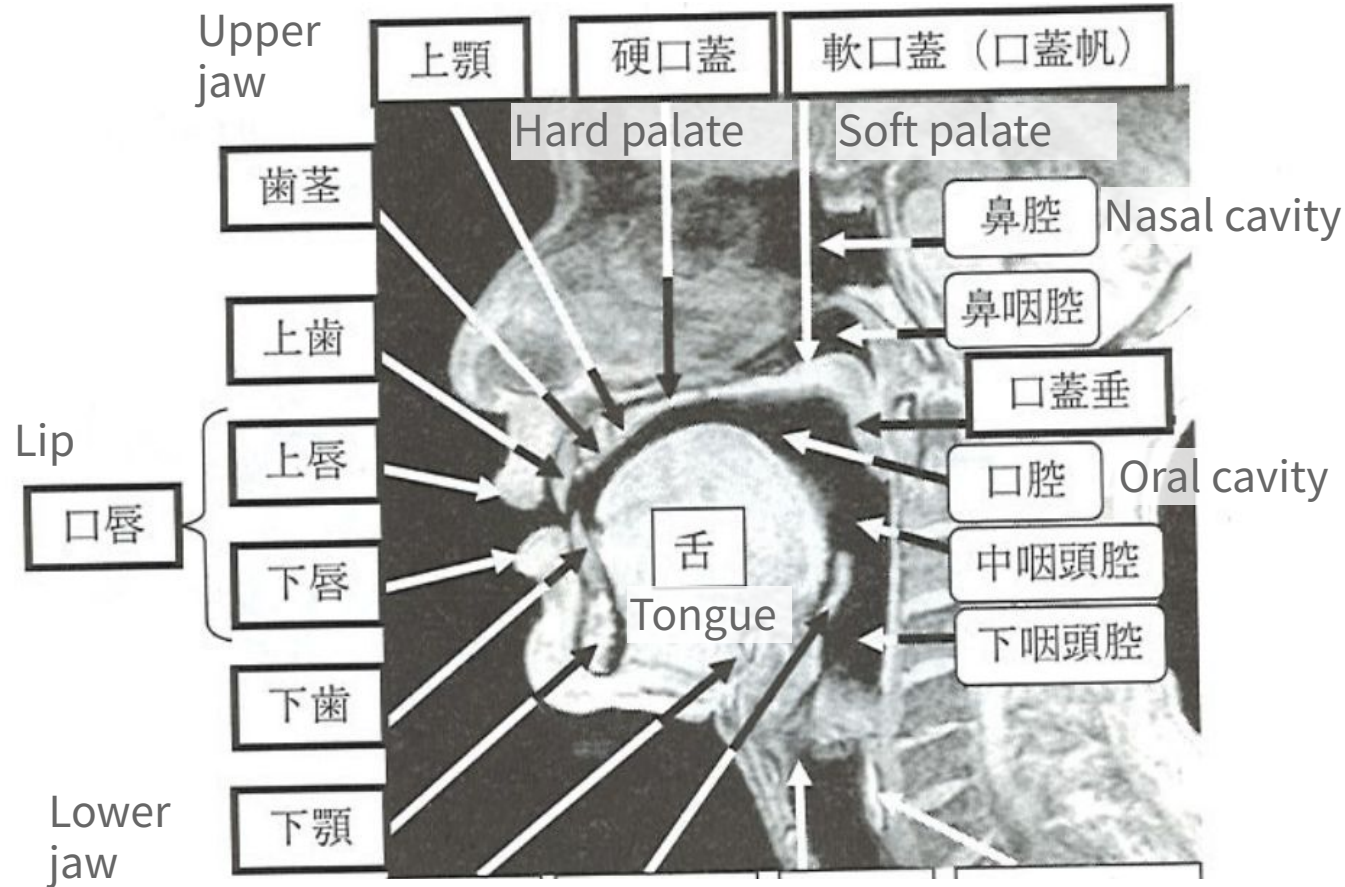
NINJAL  
フォーラム

講演

## リアルタイムMRI 動画を用いた 調音音声学の再構築 —ワ行子音の問題—

講師：前川喜久雄

# Vocal tract adds timbre to sound source. / 声道が音源に音色をつける



# Jaws shapes the oral cavity. / 顎が口腔を形作る

---

- **Upper jaw / 上顎 (じょうがく)**

- Hard palate (硬口蓋): hard parts in palate
- Soft palate (軟口蓋): soft parts in palate
  - It also serves to separate the nasal cavity from the oral cavity.
- alveolar process (歯槽突起)
- Alveolar ridge (歯茎突起, はぐきとつき): upper teeth

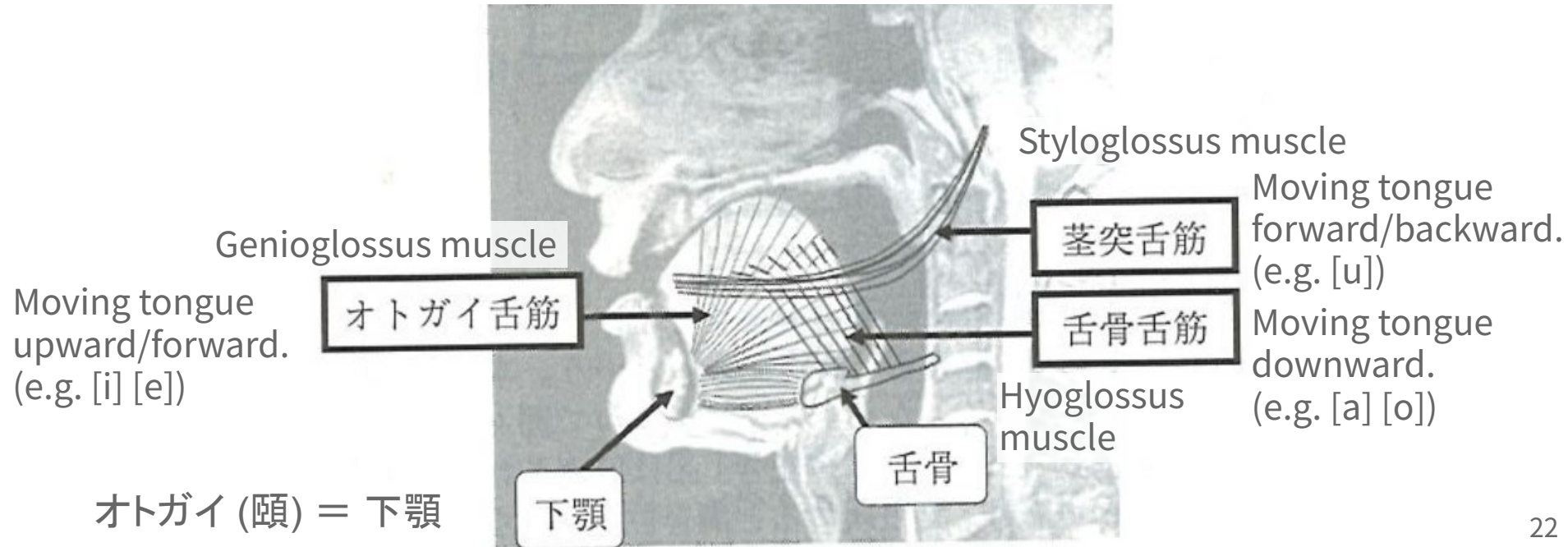
- **Lower jaw / 下顎 (かがく)**

- Rotation around the temporomandibular joint / 顎関節を軸とする回転
- Control of tongue position / 舌位置の制御

# Tongue is most important in articulation. / 舌が調音に最も寄与する

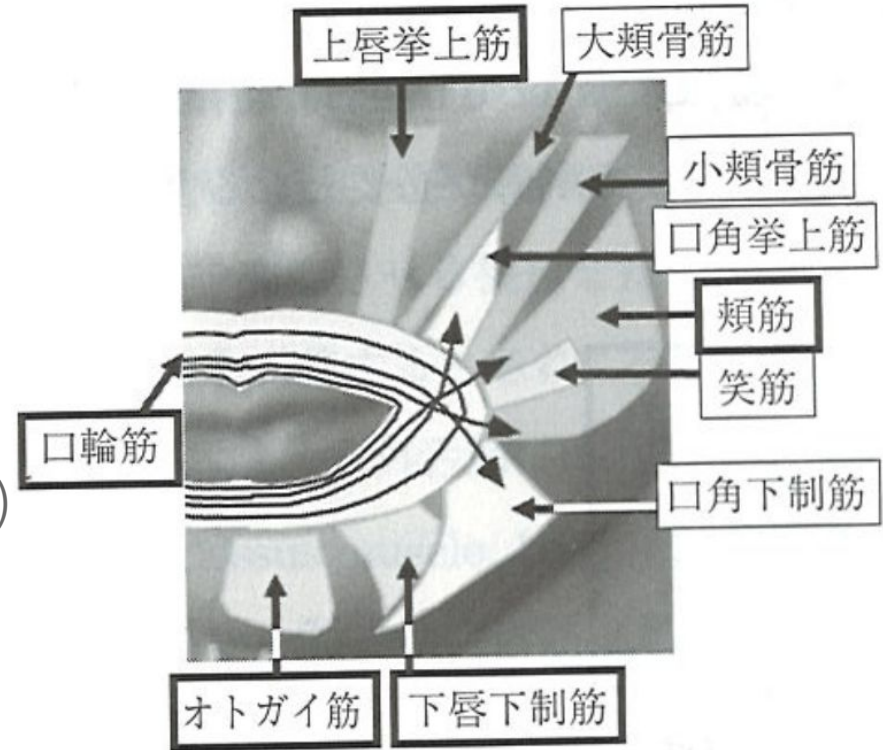
- **Extrinsic tongue muscles / 外舌筋 (がいぜつきん)**

- Originating from the surrounding bones, these muscles deform the entire tongue / 周囲の骨に起始を持ち、舌全体を変形させる



# Lips make stop, constriction, and protrusion. / 口唇は閉鎖，狭め，突き出しを行う

- Stop:
  - [p], [b], [m]
- Constriction and protrusion:
  - [ɸ] (“hu”), [w]
- Controlled by
  - orbicularis oris muscle (口輪筋)
  - and mentalis muscle(オトガイ筋)



# Source-filter theory / ソース・フィルタ理論

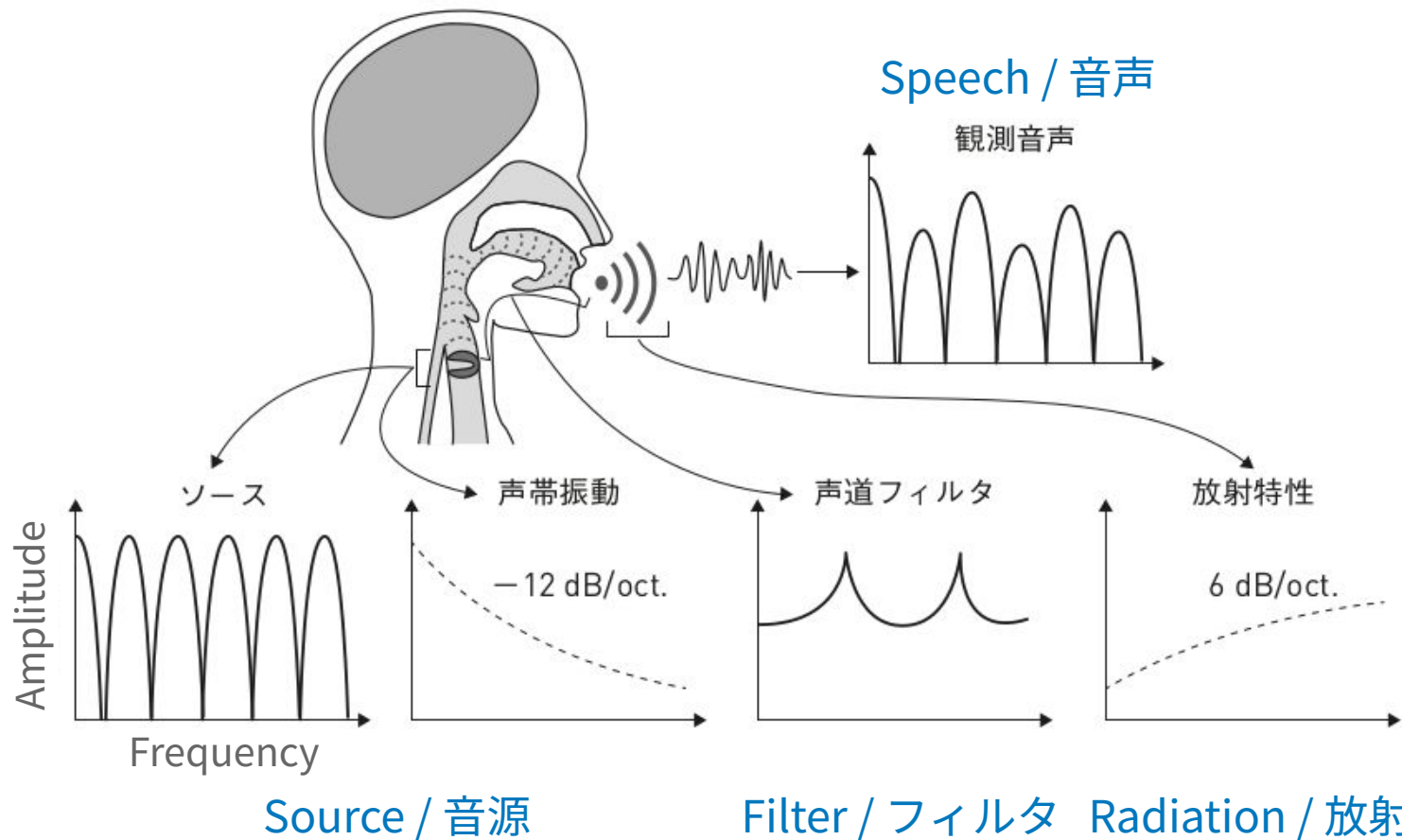


# History / 歴史

---

- Developed by Fant
  - Fant, “Acoustic Theory of Speech Production”, 1960.
  - **Cascading three linear models:**
    - Source by vocal cords / 声帯による音源
    - Filter by vocal tract / 声道によるフィルタ
    - Radiation via lip / 口唇からの放射
- Originated from Chiba (千葉) and Kajiyama (梶山)
  - Chiba and Kajiyama, “The Vowel: Its Nature and Structure”, 1941.
  - This paper introduced “production” (生成) and “articulation” (調音).

# Source-filter model / ソース・フィルタモデル



# Vowels and consonants / 母音と子音

# Vowels and consonants in linguistic sounds

## / 言語音における母音・子音

---

- Phoneme / 音素
  - Smallest units that make one word different from another.  
/ 意味の違いをもたらす, 音韻的な最小単位
- Vowel (母音, V)
  - Articulatorily: Sounds without closure involving the tongue, lips, or teeth. / 調音的には「舌, 唇, 歯での閉鎖を伴わない」音
  - Phonologically: The nucleus of a syllable that can form a syllable alone or in combination with consonants. / 音韻的には「単独あるいは子音を伴って音節を構成する」主音
- Consonant (子音, C)
  - Phonemes other than vowels / 母音以外の音

# International Phonetic Alphabet, IPA

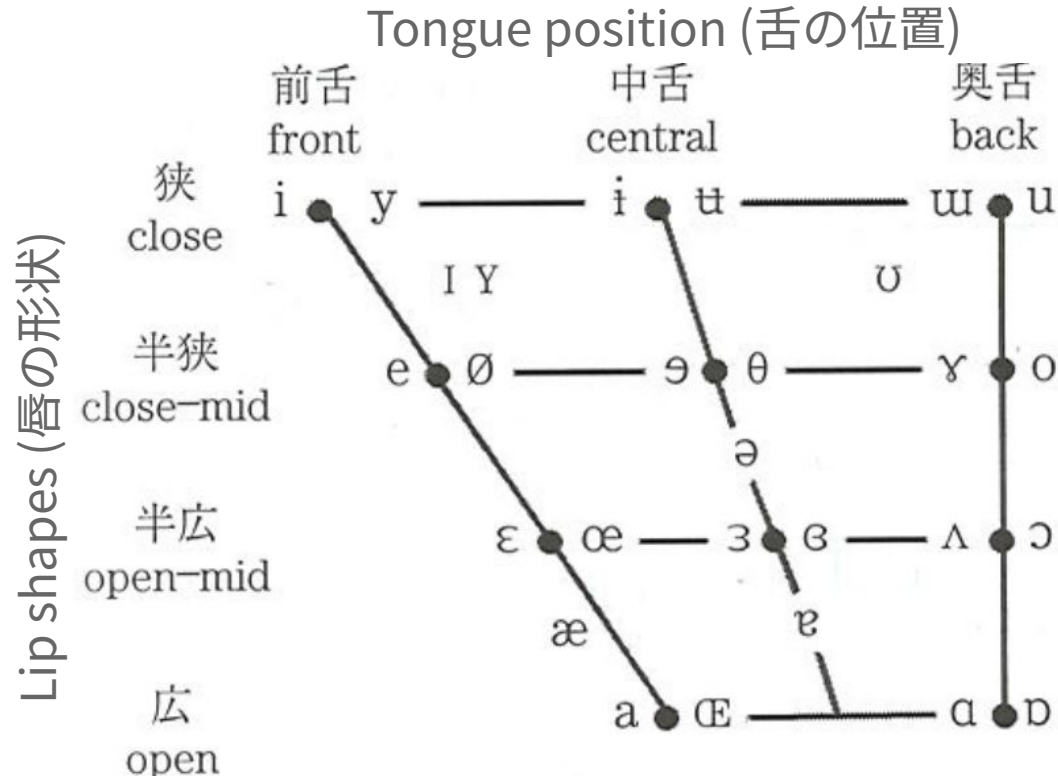
## / 国際音声記号

---

- International Phonetic Association (国際音声学会)
  - Academic society related to phonetics / 音声学に関連する学術団体.
- IPA
  - Universal symbols to describe every kinds of languages. / あらゆる言語の音声を統一的に表記するための音声記号
  - <https://www.internationalphoneticassociation.org/content/ipa-chart>

# Classification of vowels by IPA

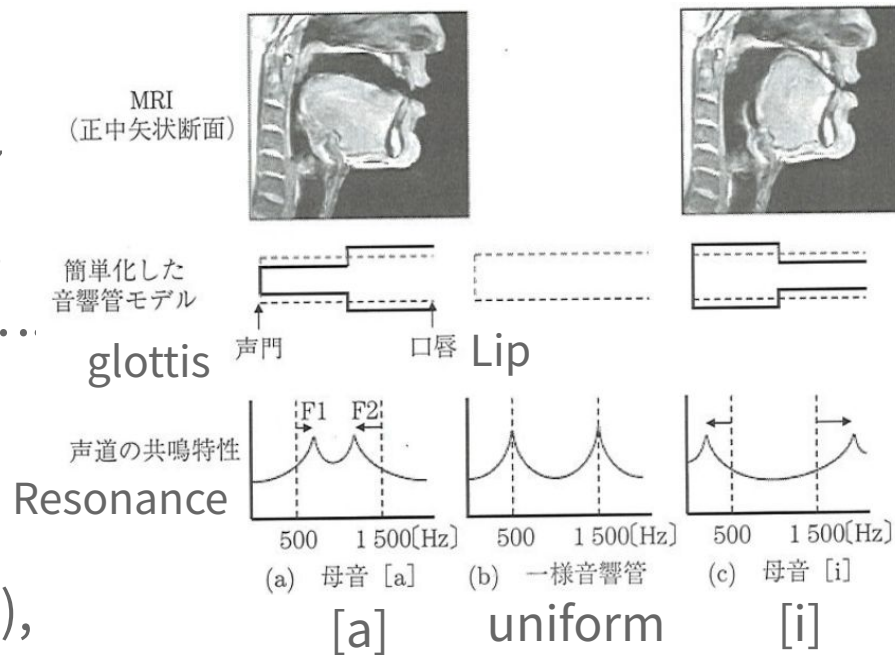
## / IPA による母音の分類



Left is unrounded lip, right is rounded lip  
左は非円唇母音, 右は円唇母音である

# Formant frequency / フォルマント周波数

- The vocal tract can be seen as concatenation of acoustic tubes.
- If the uniform acoustic tube is 17 cm long (average among males), it resonates at 500 Hz, 1500 Hz, ...
- The frequencies of these resonances are called formant frequencies (フォルマント周波数), denoted as  $F_1$ ,  $F_2$ , ... in ascending order.



# Classification of consonants by IPA

## / IPA による子音の分類

| 調音位置<br>調音様式                 | 両唇音<br>bilabial | 唇歯音<br>labiodental | 歯音<br>dental | 歯茎音<br>alveolar | 後部歯茎音<br>post alveolar | そり舌音<br>retroflex | 硬口蓋音<br>palatal | 軟口蓋音<br>velar | 口蓋垂音<br>uvular | 喉頭音<br>pharyngeal | 声門音<br>glottal |
|------------------------------|-----------------|--------------------|--------------|-----------------|------------------------|-------------------|-----------------|---------------|----------------|-------------------|----------------|
| 破裂音<br>plosive               | p b             |                    | t d          |                 |                        | ʈ ɖ               | c ɟ             | k ɡ           | q ɢ            |                   | ʔ              |
| 鼻音<br>nasal                  | m               | ɱ                  | n            |                 |                        | ɳ                 | ɲ               | ŋ             | ɴ              |                   |                |
| ふるえ音<br>trill                | ʙ               |                    | r            |                 |                        |                   |                 |               | ʀ              |                   |                |
| 弾き音<br>tap or flap           |                 | ɸ                  | ɾ            |                 |                        | ɽ                 |                 |               |                |                   |                |
| 摩擦音<br>fricative             | ɸ β             | f v                | θ ð          | s z             | ʃ ʒ                    | ʂ ʐ               | ç ʝ             | x ɣ           | χ ʁ            | ħ ʕ               | h ɦ            |
| 側面摩擦音<br>lateral fricative   |                 |                    | ɬ ɮ          |                 |                        |                   |                 |               |                |                   |                |
| 接近音<br>approximant           |                 | ʋ                  | ɹ            |                 |                        | ɻ                 | j               | ɰ             |                |                   |                |
| 側面接近音<br>lateral approximant |                 |                    | l            |                 |                        | ɭ                 | ʎ               | ʟ             |                |                   |                |

1つのセルに2つの記号がある場合、左側が無声音、右側が有声音の記号である。  
Left is unvoiced consonant, right is voiced consonants.



## Conclusion / まとめ

# Conclusion

---

- Speech organs / 音声器官
  - Respiratory organs / 呼吸器官
  - Phonatory organs / 発声器官
  - Articulatory organs / 調音器官
- Source-filter theory / ソース・フィルタ理論
  - Source by vocal cords / 声帯による音源
  - Filter by vocal tract / 声道によるフィルタ
  - Radiation via lip / 口唇からの放射
- Vowels and consonants / 母音と子音
  - IPA / 国際音声記号